Numerical Analysis Assignment 2: Round-Off and Truncation Error

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1 Introduction

In this assignment, a time-stepper was designed, implemented, and tested for the initial boundary value problem

$$u_t - \mu \sin(u) - u_{xx} = 0, \qquad -1 < x < 1, \quad t > 0,$$

with boundary and initial conditions

$$u(-1,t) = u(1,t) = 0, \qquad u(x,0) = u_0(x).$$

The linear stability of the zero equilibrium will be examined to predict the behavior of small perturbations. The equation is then discretized using an implicit-explicit Euler method in time and a centered finite difference in space. The linear system is then solved using LU factorization. Relative convergence tests will be used to check the expected first-order convergence in time and second-order convergence in space.

2 Linear Stability

Introduce a small perturbation $u = \epsilon v$ about $\bar{u} = 0$. Using $\sin(\epsilon v) = \epsilon v + O(\epsilon^3)$ and keeping only first-order terms gives

$$v_t - \mu v - v_{xx} = 0, \qquad v(-1,t) = v(1,t) = 0.$$

Substituting the test functions $v_o = e^{\lambda_o t}\sin(k\pi x)$ and $v_e = e^{\lambda_e t}\cos((k+\frac{1}{2})\pi x)$ gives

$$\lambda_o = \mu - k^2\pi^2, \quad k = 1,2,\ldots, \qquad \lambda_e = \mu - \left(k+\frac{1}{2}\right)^2\pi^2, \quad k = 0,1,\ldots.$$

The largest growth rate is $\mu - \pi^2/4$. For $\mu < \pi^2/4$, small perturbations decay, so the zero equilibrium is linearly asymptotically stable. For $\mu > \pi^2/4$, some perturbations grow, making it unstable. At $\mu = \pi^2/4$, further analysis of the nonlinear terms is needed to determine stability.

3 Numerical Methods

3.1 Discretize Time

Let $t_m = m\delta$, where δ is the time step, and let $u^m(x)$ approximate $u(x,t_m)$. Then, approximate the time derivative using

$$u_t(x,t_m) \approx \frac{u^{m+1}(x) - u^m(x)}{\delta}.$$

The implicit-explicit Euler method will evaluate the nonlinear reaction term at the current time and in the next time:

$$\frac{u^{m+1} - u^m}{\delta} - \mu\sin(u^m) - \frac{\partial^2 u^{m+1}}{\partial x^2} = 0.$$

Multiply by the δ and collect the unknown terms on the left, which gives,

$$u^{m+1} - \delta\frac{\partial^2 u^{m+1}}{\partial x^2} = u^m + \delta\mu\sin(u^m).$$

This also can be shown in operator form,

$$\left(1 - \delta\frac{\partial^2}{\partial x^2}\right)u^{m+1} = u^m + \delta\mu\sin(u^m).$$

3.2 Discretize the Solution

The first step to discretize the solution is to define a regular spatial grid by

$$x_j = -1 + jh, \qquad h = \frac{2}{n}, \qquad j = 0, \ldots, n.$$

Let u_j^m approximate $u(x_j, t_m)$, where $t_m = m\delta$ and δ is the time step. The boundary conditions give $u_0^m = u_n^m = 0$.

Use the centered second-order finite difference,

$$u_{xx}(x_j, t_{m+1}) \approx \frac{u_{j-1}^{m+1} - 2u_j^{m+1} + u_{j+1}^{m+1}}{h^2}.$$

Substitute this approximation into the implicit-explicit Euler scheme gives

$$\frac{u_j^{m+1} - u_j^m}{\delta} = \mu \sin(u_j^m) + \frac{u_{j-1}^{m+1} - 2u_j^{m+1} + u_{j+1}^{m+1}}{h^2}.$$

Multiply by δ and collect the unknown values at time t_{m+1} on the left. Which results in,

$$-\frac{\delta}{h^2} u_{j-1}^{m+1} + \left(1 + \frac{2\delta}{h^2}\right) u_j^{m+1} - \frac{\delta}{h^2} u_{j+1}^{m+1} = u_j^m + \delta\mu \sin(u_j^m), \qquad j = 1, \ldots, n-1.$$

3.3 Matrix-Vector Form

The Matrix can be written with the interior equations as

$$M = \begin{pmatrix} 1 + \frac{2\delta}{h^2} & -\frac{\delta}{h^2} & 0 & \cdots & 0 \\ -\frac{\delta}{h^2} & 1 + \frac{2\delta}{h^2} & -\frac{\delta}{h^2} & \ddots & \vdots \\ 0 & -\frac{\delta}{h^2} & 1 + \frac{2\delta}{h^2} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\frac{\delta}{h^2} \\ 0 & \cdots & 0 & -\frac{\delta}{h^2} & 1 + \frac{2\delta}{h^2} \end{pmatrix},$$

and

$$\mathbf{u}^{m+1} = \begin{pmatrix} u_1^{m+1} \\ \vdots \\ u_{n-1}^{m+1} \end{pmatrix}, \qquad \mathbf{f}^m = \begin{pmatrix} u_1^m + \delta\mu \sin(u_1^m) \\ \vdots \\ u_{n-1}^m + \delta\mu \sin(u_{n-1}^m) \end{pmatrix}.$$

The matrix M is symmetric, tridiagonal, strictly diagonally dominant, and positive definite for $\delta > 0$. It has a sparse structure because only the main and two adjacent diagonals contain nonzero entries.

These properties of matrix M allow a LU factorization without pivoting, $M = LU$, where L is unit lower bidiagonal and U is upper bidiagonal with positive entries. Since M is constant for fixed h and δ, the factorization is computed once. Additionally, each time step then requires solving $L\mathbf{y} = \mathbf{f}^m$ by forward substitution, followed by $U\mathbf{u}^{m+1} = \mathbf{y}$ by backward substitution.

4 Computational Implementation

A Python script was implemented to solve the discretized equation and test its convergence in space and time. The pseudocode outlines the main steps: initializing the variables and matrix, computing the LU factorization, and advancing the solution through time. At each time step, the right-hand side is calculated from the current solution, and the linear system is solved using forward and backward substitution. The resulting values then replace the current solution.

Additionally, two relative convergence tests compare numerical solutions at the same final time. For time convergence, the time step δ is repeatedly halved while the spatial grid remains fixed, with first-order convergence expected. For spatial convergence, the grid spacing h is repeatedly halved while a small time step remains fixed, with second-order convergence expected. Solutions on different spatial grids are compared at shared grid points. In both tests, successive solution differences are used to estimate the observed convergence order.

4.1 Pseudocode

Algorithm 1 Time-stepper

Require: Initial interior values $\mathbf{u}$, parameter μ, number of intervals n, time step δ, final time t_f

$h \leftarrow 2/n$

$N_t \leftarrow t_f/\delta$ ▷ Assume this is an integer

$r \leftarrow \delta/h^2$

Initialize the $(n-1) \times (n-1)$ matrix M with diagonal $1+2r$ and off-diagonals $-r$

Factor $M = LU$

for $m = 0, \ldots, N_t - 1$ **do**

  for $j = 1, \ldots, n-1$ **do**

    $f_j \leftarrow u_j + \delta\mu \sin(u_j)$

  end for

  Solve $L\mathbf{y} = \mathbf{f}$ by forward substitution

  Solve $U\mathbf{u}_{\text{new}} = \mathbf{y}$ by backward substitution

  $\mathbf{u} \leftarrow \mathbf{u}_{\text{new}}$

end for

return $\mathbf{u}$, with zero boundary values

4.2 Results

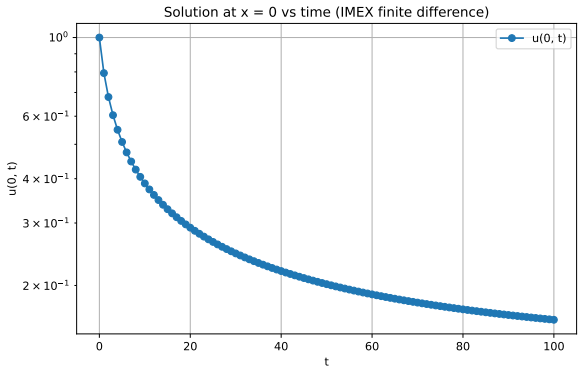


Figure 1: Solution at $x = 0$ over time.

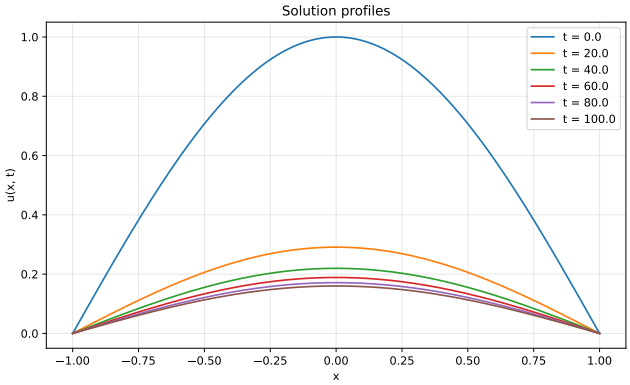


Figure 2: Solution profiles at selected times.

Figures 1 and 2 show that the solution height and amplitude decreases rapidly at first, then more slowly, reaching $u \approx 0.16$ at the end. Notably the spatial profiles remain symmetric about $x = 0$ and satisfying the zero boundary conditions throughout all the values of t.

Although $\mu > \pi^2/4$, zero being unstable does not mean the solution must always grow. The stability analysis applies near zero, while this solution starts with an amplitude of one. A longer simulation would help show whether it settles at a nonzero value.

4.3 Convergence Tests

Both tests used $\mu = 1$, $u_0(x) = \cos(\pi x/2)$, and a final time of $t_f = 0.5$. Solution differences were measured using the maximum absolute difference at shared spatial grid points.

4.3.1 Time Convergence

For this test $n = 200$ and δ was repeatedly halved from 0.05. Successive solutions were compared using

$$D_\delta = \|\mathbf{u}_\delta(t_f) - \mathbf{u}_{\delta/2}(t_f)\|_\infty, \qquad p_\delta = \log_2(D_\delta/D_{\delta/2}).$$

First-order convergence gives $p_\delta \approx 1$: halving δ approximately halves the solution difference.

In Figure 3, the solution differences follow the first-order reference, supporting spatial convergence of order $O(\delta)$.

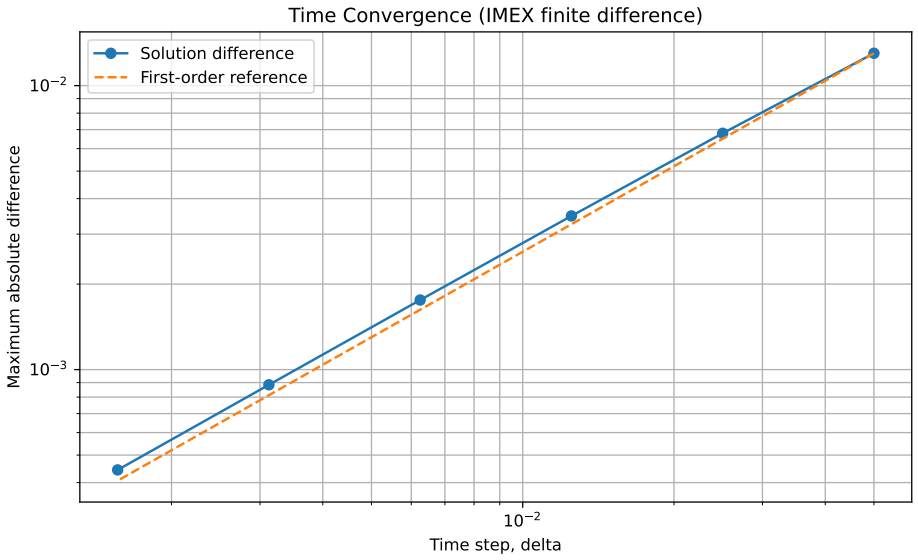


Figure 3: First order reference compared to the solution difference on a log-log plot.

4.3.2 Space Convergence

With $\delta = 0.0005$ fixed, n was doubled from 10 to 160. Solutions were compared at shared grid points using

$$D_h = \|\mathbf{u}_h(t_f) - \mathbf{u}_{h/2}(t_f)\|_\infty, \qquad p_h = \log_2(D_h/D_{h/2}).$$

Second-order convergence gives $p_h \approx 2$: halving h reduces the solution difference by approximately four.

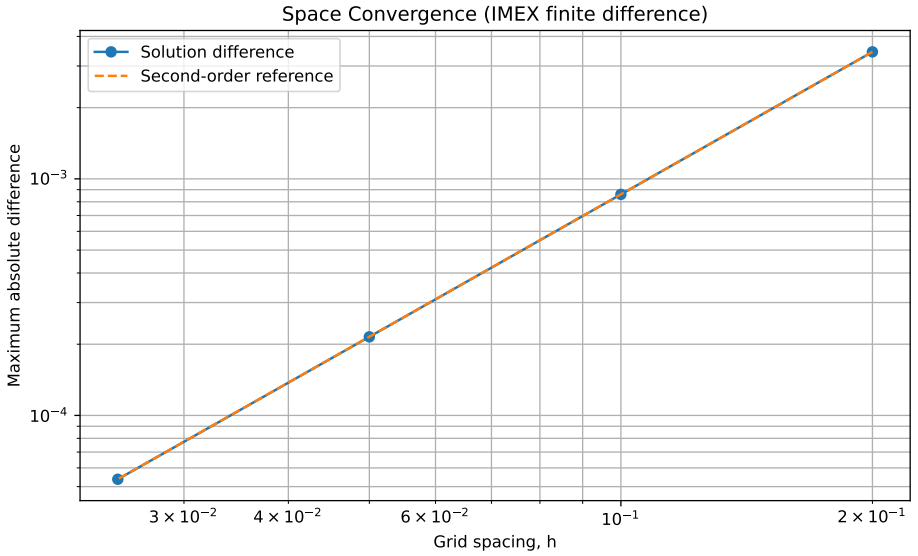


Figure 4: Second order reference compared to the solution difference.

In Figure 4, the solution differences follow the second-order reference, supporting spatial convergence of order $O(h^2)$.

5 Conclusion

An implicit-explicit Euler time-stepper was implemented using a centered spatial finite difference and LU factorization. The relative convergence tests supported first-order convergence in time and second-order convergence in space

for the tested parameters and initial condition. The linear stability analysis identified $\mu = \pi^2/4$ as the threshold separating decay and growth of small perturbations about the zero equilibrium.